AGE 316: MICROCOMPUTER SYSTEMS

LECTURE 3: ELECTRONICS & DIGITAL I/O

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Recap: Lecture 2

- **ATmega328P:** The brain of the Arduino Uno.
- **Memory:** Flash (Code) vs. SRAM (Variables).
- **I/O Pins:** Digital (0-13) and Analog (A0-A5).
- **Power:** 5V USB or 7-12V DC Jack.

Ready to power up?

Understanding Electricity

Before we can program the microcontroller, we must understand the **flow of energy** that powers it.

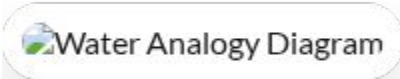


Voltage **Current** **Resistance**

The Water Tank Analogy

Think of electricity like water flowing through a pipe:

- **Voltage (V):** Water **Pressure**. The "push" from the tank.
- **Current (I):** Water **Flow Rate**. How much water is moving.
- **Resistance (R):** Pipe **Width**. A thin pipe resists flow more than a wide one.



The Golden Rule: Ohm's Law

The relationship between Voltage, Current, and Resistance is defined by:

$$V = I \times R$$

V (Volts)
The potential difference pushing the electrons.

I (Amperes)
The rate of electron flow.

R (Ohms Ω)
How much the material fights the flow.

Component 1: The Resistor



Function

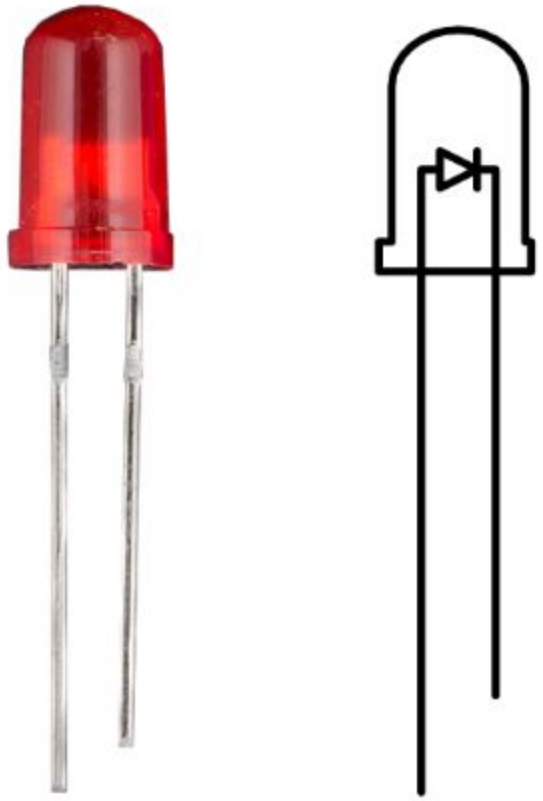
Limits the flow of current in a circuit. Used to protect sensitive components (like LEDs).

Identification

Values are marked with **Color Bands**.

Unit: Ohm (Ω)

Component 2: The LED



Light Emitting Diode

A semiconductor that emits light when current flows through it.

Polarity Matters!

Anode (+): Long leg. Connect to positive voltage.

Cathode (-): Short leg/Flat side. Connect to Ground (GND).

Component 3: The Push Button



Tactile Switch

A simple mechanical switch that closes a circuit when pressed.

Function

Used as a **Digital Input** device. It tells the Arduino when a user has interacted with the system.

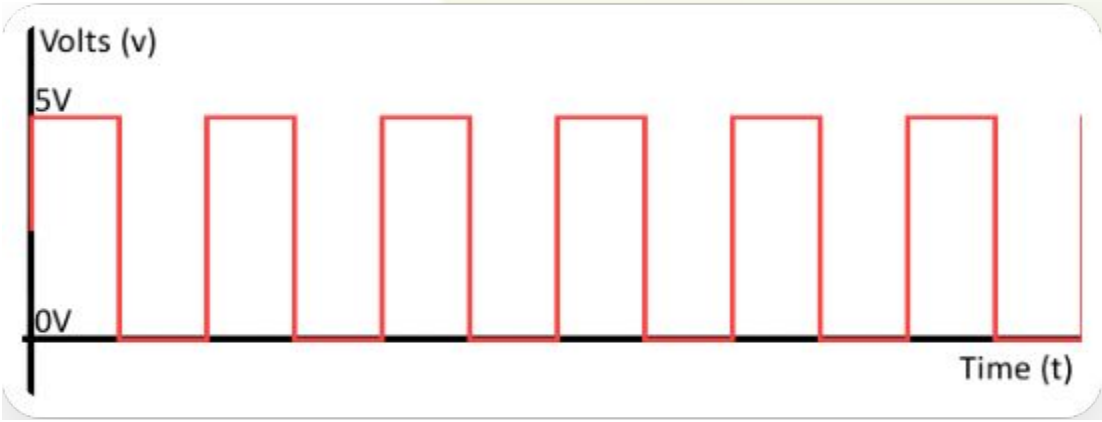
State: Open (Disconnected) vs. Closed (Connected).

Digital vs. Analog Logic

The Binary World

In digital electronics, there are only two states:

HIGH	LOW
1	0
5 Volts	0 Volts (GND)



A Digital Signal (Square Wave)

Arduino Function: pinMode()

Configures a specific pin to behave either as an input or an output.

```
void setup() {
  // Configure pin 13 to drive an LED
  pinMode(13, OUTPUT);

  // Configure pin 2 to read a button
  pinMode(2, INPUT);
}
```

Note: This is usually done once in the setup() function.

Arduino Function: digitalWrite()

Write a **HIGH** or a **LOW** value to a digital pin. Used to turn things On or Off.

```
void loop() {
  digitalWrite(13, HIGH); // Turn LED ON (5V)
  delay(1000);
  digitalWrite(13, LOW); // Turn LED OFF (0V)
  delay(1000);
}
```

Arduino Function: digitalRead()

Reads the value from a specified digital pin, either **HIGH** or **LOW**.

```
int buttonState = 0;

void loop() {
  // Read the state of the button
  buttonState = digitalRead(2);

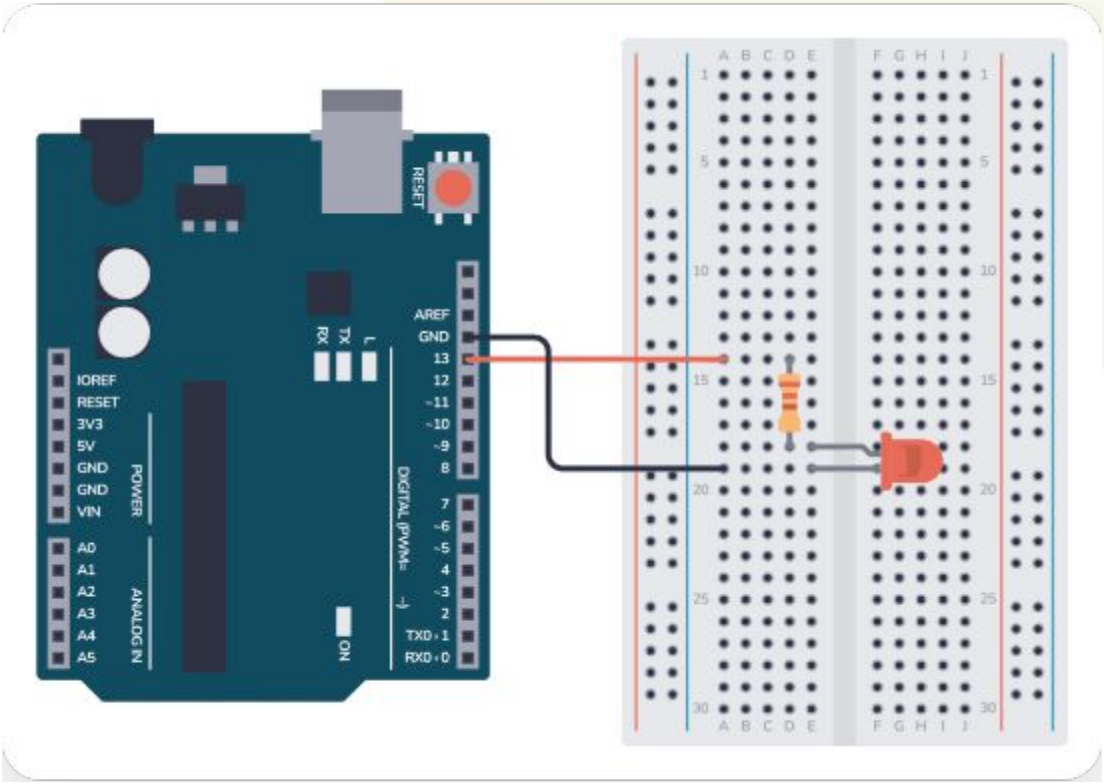
  if (buttonState == HIGH) {
    // Button is pressed...
  }
}
```

Project: The "Blink" Sketch

The "Hello World" of hardware. This code makes an external LED blink on and off.

```
void setup() {
  pinMode(8, OUTPUT);
}

void loop() {
  digitalWrite(8, HIGH);
  delay(1000);
  digitalWrite(8, LOW);
  delay(1000);
}
```



Concept: Floating Pins

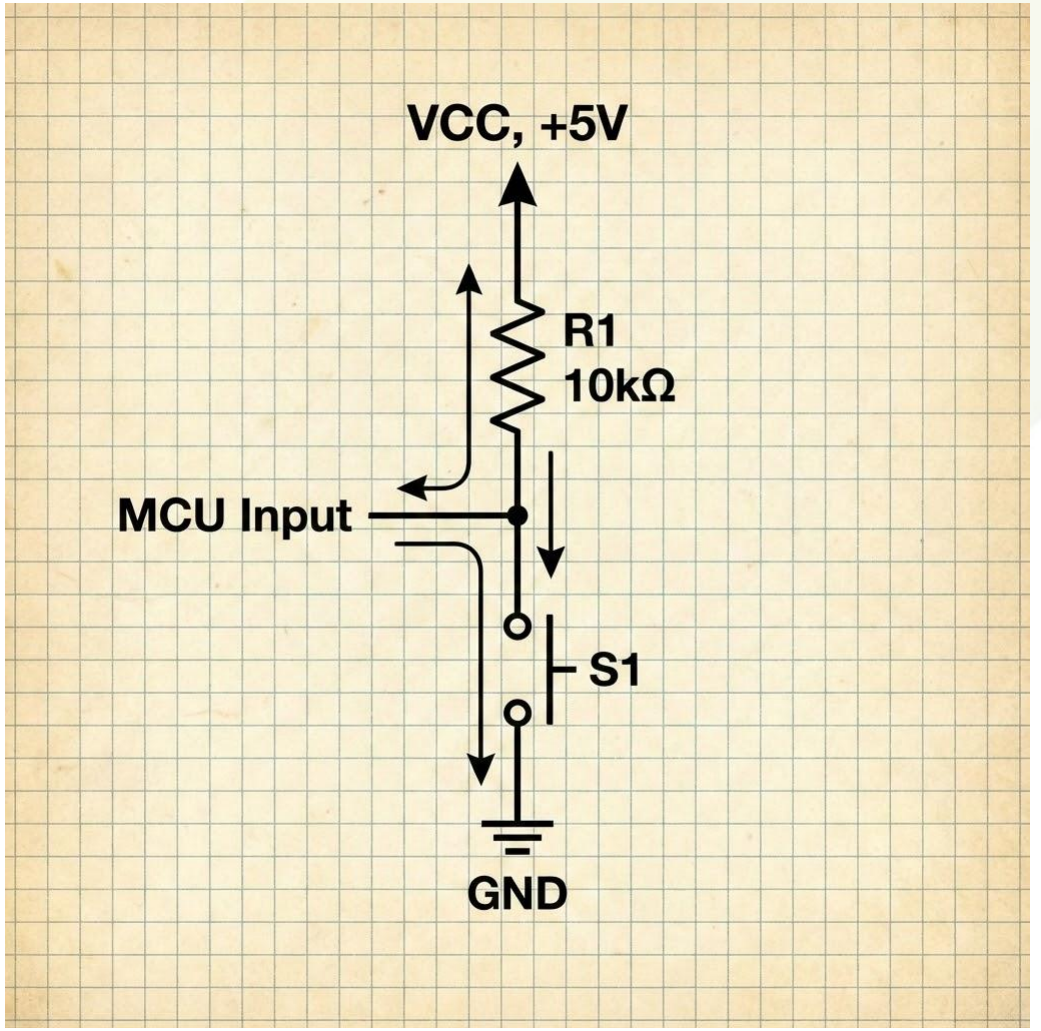
The Problem

When a switch is open, the input pin is not connected to anything. It is **"Floating"**.

It can randomly read HIGH or LOW due to electrical noise.

The Solution

Use a **Pull-up Resistor** (usually 10kΩ) to pull the pin to 5V when the switch is open.



Summary



Basics

Voltage, Current, and Resistance (V=IR) govern all circuits.



Components

Resistors limit flow. LEDs emit light. Buttons provide input.



Logic

pinMode, digitalWrite, and digitalRead are the core I/O functions.

Lab Preparation

Bring your **Arduino Kit** (Board, USB, Breadboard, Wires, LEDs, Resistors).

Have **Arduino IDE** installed.

We will build the **Blink Circuit** physically in the lab!

Questions?

"The only stupid question is the one you don't ask."

Dr. Kadeghe Fue

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Image Sources



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Source: wikipedia.org

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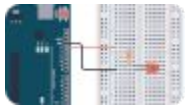
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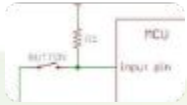
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Image Sources



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